Smart Insole: Stand-alone Soft 3-axis Force Sensing Array in a Shoe

Jian Gao[1], Zihao Pu[1], Ruixin Qiu[1], Ying Li[1], Xiulun Yin[1], Kieran Morton[1], Sadan Wani[1], Justin Wyss[1], Michael Steszyn[2], Ryusuke Ishizaki[3], Fumiya Hamatsu[3], Takeshi Ohsato[3], John D.W. Madden[1,*]

[1] Department of Electrical & Computer Engineering, Advanced Material Processing Engineering Laboratory, University of British Columbia, Vancouver, Canada

[2] i-generator, Portland, USA

[3] Frontier Robotics, Honda R&D Co., Ltd, Saitama, Japan

*Email: jmadden@ece.ubc.ca

***Abstract*—Foot plantar force monitoring has been explored using in-shoe soft insole measurements for application in diabetic foot and to assess footwear performance. Sensors are impractical for regular use due to bulky electronics and short battery life. They typically lack shear sensing. A soft flexible and custom configurable 3-axis capacitive sensing array is demonstrated through a customizable arrangement of 16-shear and normal force taxels completely imbedded in a shoe insole that wirelessly updates at 50 Hz. A patterned silicone dielectric mounted on a flex printed circuit board enables normal and shear sensitivities of 19.9 and 2.8 kPa^{-1}. The 20 mW power consumption includes transmission of data to a nearby mobile device, which is satisfied for more than 50 hours using a built-in 350 mAh LiPo battery. Operation in a shoe is demonstrated.**

Keywords—Wearables, Capacitive sensing, Pressure sensor, Shear sensor, Health monitoring

I. Introduction

Foot plantar force analysis is essential for a wide range of applications including footwear design [1], [2], sports performance analysis [3], [4], injury prevention [5], [6], rehabilitation support systems [7], [8], and biometric identification [9], [10]. In these studies, external force platforms are the standard. In-shoe measurement is often preferred because of the flexibility, mobility, reduced costs, and ability to test continuously in a realistic setting [11]. However, such systems have limited battery life, along with electronics and power external to the shoe, making them unsuitable for regular and unencumbered use.

To provide in-shoe measurement, several sensors have been implemented. Khandakar *et al.* [12] implemented Force Sensitive Resistors (FSRs) for 16 normal-force taxels consuming 400 mW of power. The FScan Go is a commercially available array with 4 normal force sensing elements/cm^2 and a thickness of 0.3 mm. This system comes with ankle mounted electronics and battery, or in a 64 element clinical sensor (0.6/cm^2) with a 27 g electronics box outside the shoe, 50/100 Hz sample rate, 10 m range, and 1.5-hour battery life [13]. XSENSOR is another commercial system with similar normal force resolution, based on capacitive sensing, equipped with 46 g ankle mounted electronics scanning for 8 hours [14]. Tang *et al.* [15] recently created the first flexible insole sensor for both normal force and shear - the latter a major risk factor for diabetic ulcer formation (1 in 3 people with diabetes develop these sores, and of those, 1 in 4 have foot amputations). Shear is also necessary for producing acceleration in walking and running. To detect gait events for diabetic patients, 3-axis taxels were positioned at 4 locations where diabetic foot ulcers often occur. All the insole systems to date require connection to external microcontrollers and batteries, consume substantial power, and have no or low resolution shear detection.

This paper presents a smart material insole solution with all electronics integrated into a US size 9 insole. Force and time resolution were chosen to match those of commercially available clinical systems, but with a shear measurement array, as shear induced sores can occur anywhere on the foot. A reconfigurable sensing array is used to implement 16, 3-axis force taxels located at 3 high pressure zones, and 80 force sensing locations. It demonstrates the potential for day-to-day applications such as continuous pressure ulcer monitoring and gait tracking. The key to its effectiveness is the unique shear sensing array design.

II. Dimensions

A. Taxel to Measure Normal and Shear Stress

Each normal and shear sensing taxel is constructed as shown in Fig. 1. Similar to previous work [16]–[18], each taxel includes a 10 x 10 mm^2 excitation (Tx) electrode and five 4 x 4 mm^2 receiver (Rx) electrodes, forming a 5-channel capacitive sensing scheme. The elastomer pillar dielectric enables greater force sensitivity relative to a continuous dielectric. Each taxel is 14 x 14 x 2.5 mm^3 (W x L x H).

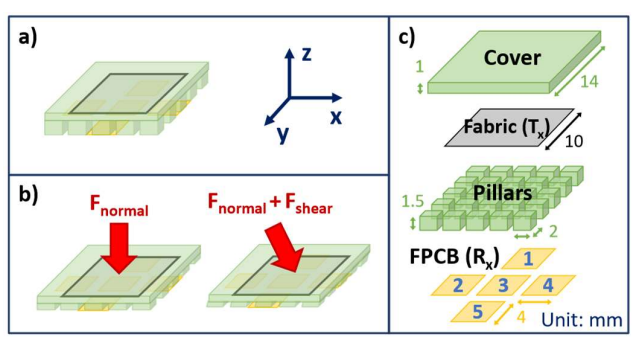

Fig. 1. a) Smart material insole's taxel under no force, b) taxel under normal and shear forces, and c) exploded view of the taxel.

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A parallel plate capacitor model, $C = \varepsilon A/d$, is used to separate normal force from shear in each element, where ε represents the permittivity; A represents the overlap area between two plates; and d represents their separation distance. By applying a z-axis stress uniformly on the top of the taxel, the elastomer structure deforms so d decreases. The total capacitance (C_{all}) between Tx and 5 Rx electrodes increases in inverse proportion to d. When a shear stress is introduced, the capacitances of the 4 side channels (1,2,4,5) change based on individual overlap area. Thus, the planar displacement of the Tx electrode is estimated according to the capacitance differences in opposing channels ($\frac{C_4-C_2}{C_4+C_2}$ for x-axis; $\frac{C_1-C_5}{C_1+C_5}$ for y-axis), leading to good estimation of shear magnitude and direction [17]. Unique to this work is the demonstration of these taxels in a scalable array.

B. *Sensing Array for Toe, Forefoot, and Heel*

Based on normal force heatmaps [15], taxels are arranged into a 3 rows x 1 column taxel array designed for the toe, alongside 3 x 3 and 2 x 2 arrays designed for the forefoot and heel. Common electrical connections are made for each column of Tx electrodes (6 total), and there are 5 connections for each row of Rx electrodes (5 rows x 5 = 25).

To manage the 31 connections, a flexible printed circuit board (FPCB) is adopted. A flexible printed circuit (FPC) header connects to a custom capacitive sensing circuit. The FPCB is shaped to a US men's size 9 insole. A feature of the array design is that sizing can readily be changed by cutting (or by printing custom FPCBs, $1 each in small quantities).

III. Fabrication

For smart insole, Ecoflex™ 00-30 (Smooth-On) was selected as the elastomer to encapsulate the Tx electrodes and to act as the dielectric as it bonds readily to the FPCB with a thin layer of SS-6000 adhesive (Silicone Solutions) and deforms sufficiently in the force range of interest to produce readily observable changes in capacitance. It can be replaced by other elastomers to tune the sensitivity for different needs (e.g. stiffer for roller automate fibre placement smart roller [19]). To form the Tx structure with the cover and pillars seen in Fig. 1, a mould was designed for a 6 x 6 sensing array and 3D printed by a Prusa 3S printer using PLA. 10 g of Ecoflex™ (5 g of Part A/ B) was poured into the mould and degassed. Left under the ambient temperature for 4 hours, the first layer of Ecoflex™ cured and pillars were formed. Second, the Tx electrodes were made by laser cutting stretchable silver-coated conductive fabric (LessEMF) and aligned to the mould. Third, 15 g of Ecoflex™ was poured on top of the Tx electrodes for encapsulation and degassed. After 4 more hours under ambient temperature, the Ecoflex™-housed 6 x 6 Tx electrodes were extracted from the mould and cut into two sets of 3 x 1, 3 x 3, and 2 x 2 arrays, as depicted in Fig. 2. They were attached to factory-made FPCBs for two insoles.

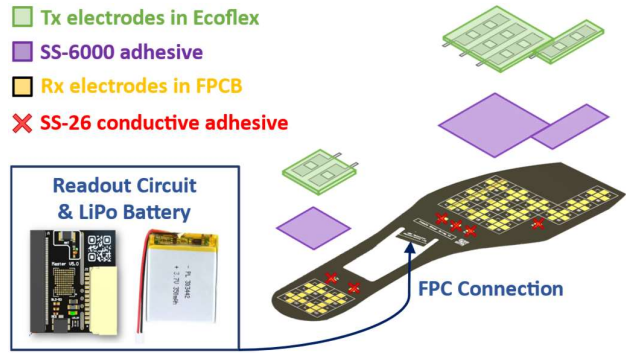

Fig. 2. Assembly of the soft flexible smart insole.

An 8-layer custom readout circuit was developed around a Cypress PSoC63 MCU at a size of 13 x 23 x 4 mm^3 (W x L x H). By utilizing the MCU's built-in BLE and CAPSENSE™ capacitive sensing, a low power profile was achieved at 20 mW. The circuit was installed to the insole alongside a 350 mAh 3.7 V LiPo battery. As the sensor's only rigid components, they are situated under user's foot arch, and are not noticeable (based on one user test).

IV. System Evaluation

A. *Wireless Data Collection via BLE*

To collect and visualize capacitance data from the sensor on the run via BLE 5.0, a custom mobile application [20] was crafted for iPhones/iPads/Apple Watches (iOS/iPadOS 15.0+, watchOS 9.0+) using Swift 5. Shown in Fig. 3, both normal and shear forces detected by 16 taxels are rendered at 50 Hz. For normal force, each taxel's color represents the normalized value of the combined increase in C_{all}. For the shear force, an arrow indicator was implemented with its size and orientation based on the normalized estimated planar displacements. With no normal or shear force, the taxel is demonstrated as a blue square (left) and no arrow (right).

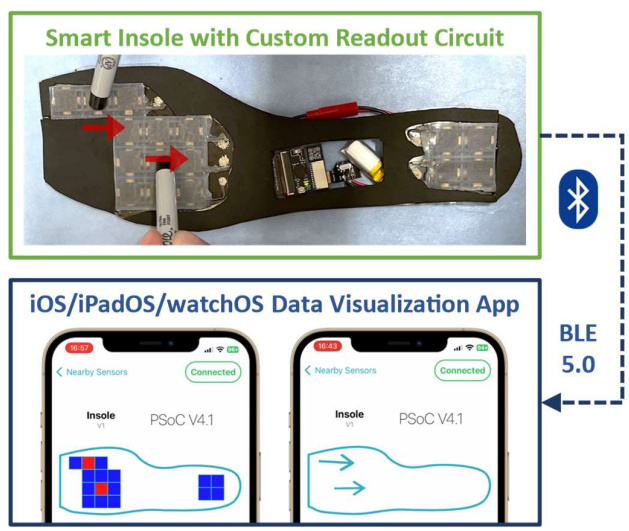

Fig. 3. Collection (top) and display for normal (left) and shear (right) forces where no crosstalk is noticed in the adjacent taxels.

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B. *Characterization for Normal Stress and Shear*

For normal force characterization, a 0-0.8 mm step sequence (0.2 mm and 5 s per step) was applied to the central forefoot taxel using an Instron 5969 Universal Testing Machine. Fig. 4 reveals the approximate linearity (slope of 19.9 kPa^{-1}) between the taxel's normalized C_{all} and the normal stress, for indentation $\leq$ 0.6 mm. Non-linear performance is noted for indentation $\geq$ 0.8 mm. The z-axis stress is calculated from the recorded force and the indenter's contact area (14 x 14 mm^2), whereas the normalized delta C_{all} value is a measurement given by the custom readout circuit, proportional to the change in raw capacitance. The viscoelasticity of Ecoflex™ and SS-6000 is the likely cause of the relaxation and lower stress during unloading.

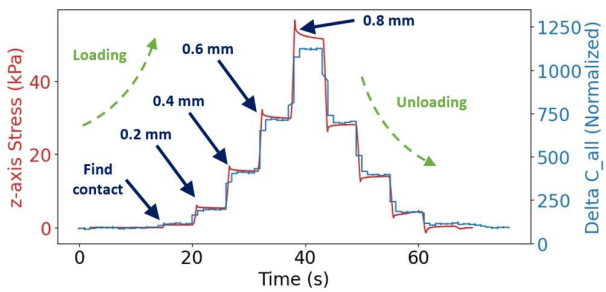

Fig. 4. Normal stress step response.

A 3-axis actuation setup was used for shear taxel characterization (Fig. 5). It incorporates a NanoMax 300 3-axis stage, an ATI SI-50-0.5 3-axis load cell and a BoliOptics SG02201211 lift. This setup enables precise displacement in all three axes and measures forces up to 10 N with a resolution of 0.0125 N. To prevent the slippage between the taxel and the load cell, a 3 N normal force was first applied. Then, a 0-1.8 mm step sequence (0.2 mm and 10 s per step) was configured for the positive and negative directions of the x and separately of the y axis. Fig. 6 illustrates the taxel's shear response in x and y directions. During the characterization, a maximum shear stress of 15 kPa was applied, comparable to the range described by Tang *et al.* [15]. With little crosstalk between shear axes, the taxel exhibited similar sensitivity in detecting x and y-axis shear stress (slopes of 2.81 and 2.83 /kPa). The responses shown in Fig. 6 are typical of each taxel, and consistent within 10%. For precise measurements (within 5%) each taxel needs to be calibrated. Previous work has shown little cross-talk between shear and normal force measurements [17], [18].

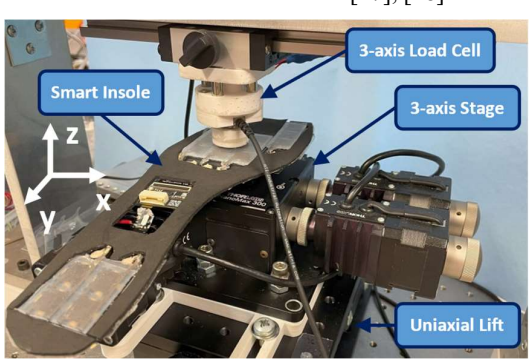

Fig. 5. Characterization setup for normal and shear stress.

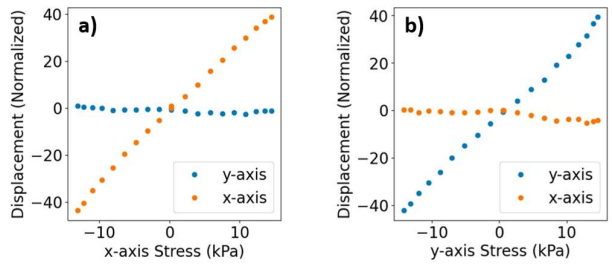

Fig. 6. Shear characterization applied in a) x-axis b) y-axis.

V. Demonstration

With characterization done, the smart insole is ready to estimate the 3-axis stress received at all 16 taxels. Fig. 7 shows the normalized forces when the author (183 cm; 82 kg), walks straight in a trainer embedded with the smart insole. The average measurement of forefoot/heel taxels is shown respectively after applying a moving average filter (window = 10 samples). As it resembles forefoot's result, the measurement of 3 toe taxels is omitted in Fig. 7.

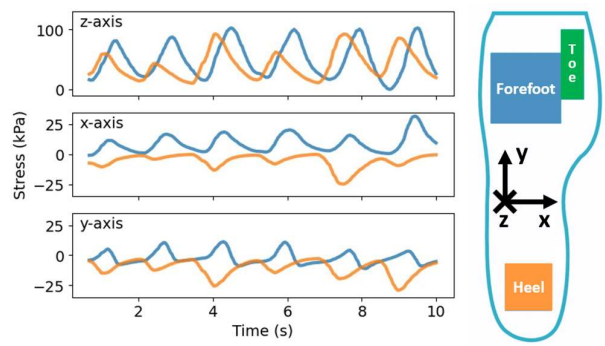

Fig. 7. Measured forefoot/heel 3-axis stress during a walk (6 steps).

During the recorded steps, a consistent stress pattern is seen in all three axes—forefoot's peak proceeds heel's peak in each gait cycle. In z-axis, the maximum stresses are 110 kPa at forefoot and 96 kPa at heel. In x-axis, forefoot and heel encounter opposing shear stress (33/-25 kPa max). The forefoot's y-axis shear alternates between positive and negative directions (13/-13 kPa max) while heel shows negative shear (30 kPa max). The shear measurements match the ranges found in [15] at metatarsal (forefoot) and heel locations. The z-axis stress at both the forefoot and heel is non-zero even when the foot is off the ground likely due to the tightness of the laces maintaining force on the insole.

VI. Conclusion

This paper demonstrates a low power wireless and soft 3-axis force sensor array applied in a shoe insole. It gives consistent stress readings among gait cycles. Directions of future work include calibration of all taxels, fabrication with stiffer dielectrics for higher force range, deployment of smart insoles under both feet to examine imbalances (typical in most people [21]), and further analysis on the 3-axis stress data at different taxels for a comprehensive understanding of the user's motion.

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